

AI-BFSI Internship Project Documentation

Problem–Solution Fit – Subscription Waste Detector

Cover Page

Title: Problem–Solution Fit **Subtitle:** Subscription Waste Detector **Program:** AI Specialist BFSI Program **Purpose:** This document explains how the implemented features of the Subscription Waste Detector directly address the identified financial problem and satisfy the project objectives.

Executive Summary

Problem–Solution Fit validates whether a software solution effectively addresses the identified problem. For the Subscription Waste Detector, the primary challenge is the lack of visibility into recurring subscription expenses.

Users often subscribe to multiple digital services, making it difficult to monitor payments, identify unnecessary subscriptions, and understand monthly spending.

The implemented application addresses these challenges through:

- Subscription detection
- Expense visualization
- Renewal reminders
- AI-powered cost-saving recommendations (via Groq API)

This ensures that raw expense data is transformed into actionable financial insights.

Introduction

Software projects must solve real user problems to deliver value. The Subscription Waste Detector was designed to address the growing challenge of managing subscription-based digital services.

By focusing on recurring subscription analysis, the system improves financial awareness and empowers users to make informed decisions about their expenses.

Problem Overview

Consumers increasingly rely on subscription services (streaming, cloud storage, productivity tools, education, AI platforms). While individually affordable, combined costs often become significant.

Automatic renewals mean users frequently pay for unused or forgotten services. Traditional expense trackers show transactions but rarely highlight recurring subscriptions or provide intelligent recommendations.

Identified Problems

- **Problem 1 — Limited Visibility of Recurring Expenses** Users cannot easily track monthly subscription spending.
- **Problem 2 — Forgotten Subscription Renewals** Automatic renewals continue charging users for unused services.
- **Problem 3 — Manual Transaction Analysis** Reviewing transaction histories manually is time-consuming.
- **Problem 4 — Lack of Intelligent Financial Guidance** Existing tools provide records but limited actionable insights.

Implemented Solution

The Subscription Waste Detector solves these problems by:

- Allowing CSV/Excel uploads and manual entry.
- Using Python + Pandas to detect recurring subscriptions.
- Presenting results in an interactive Streamlit dashboard with Plotly charts.
- Generating AI-powered recommendations via Groq API.
- Providing renewal reminders and subscription listings.

Problem–Solution Mapping

Identified Problem	Implemented Solution
Limited visibility of subscription expenses	Dashboard with detected subscriptions and monthly spending charts
Forgotten recurring subscriptions	Renewal reminder functionality
Manual review of transactions	Automated recurring subscription detection
Lack of financial insights	AI-powered cost-saving recommendations via Groq API
Difficulty understanding spending patterns	Interactive charts and expense visualization

How the Solution Fits the Problem

The solution is narrowly focused on subscription analysis rather than general expense management. This targeted approach ensures:

- Direct visibility into recurring payments.
- Actionable insights through AI recommendations.
- Simplified monitoring via dashboards and charts.

By leveraging Groq API, the system provides intelligent suggestions without requiring users to interpret raw transaction data.

User Benefits

- **Financial Awareness:** Clear visibility into recurring expenses.
- **Reduced Unnecessary Spending:** Identification of unused subscriptions.
- **Better Decision Making:** AI-generated recommendations.
- **Improved Monitoring:** Interactive dashboards simplify analysis.
- **Time Savings:** Automated detection replaces manual review.

Project Scope Alignment

Included:

- CSV/Excel upload
- Manual entry
- Subscription detection
- Dashboard visualization
- Plotly charts
- Renewal reminders
- AI-powered recommendations via Groq API

Excluded (Future Enhancements):

- Authentication
- Database storage
- Bank API integration
- Investment/loan/tax management
- Budget planning
- Multi-user support

Validation of Problem–Solution Fit

Every implemented feature directly addresses one or more identified problems. The solution avoids unnecessary complexity and remains focused on subscription expense analysis.

This alignment ensures the project is relevant, technically feasible, and valuable to end users.

Conclusion

The Subscription Waste Detector demonstrates strong **Problem–Solution Fit** by directly addressing the challenges of subscription expense management.

Through recurring subscription detection, expense visualization, renewal reminders, and AI-powered recommendations, the system improves financial awareness and empowers users to make informed decisions.

The project scope is respected, and the solution remains practical, defensible, and aligned with the objectives of the AI Specialist BFSI Program.

